NOT MEASUREMENT SENSITIVE

MIL-P-85856/33
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS CHROMIUM CARBIDE, TYPE II, CLASS 2, COMPOSITION B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe	C	Si	Cr
0.7 max.	12.75 min.	0.10 max.	86.0 min.

Particle Size/Distribution

+140 Mesh (106 µm)	1.0 min.
+170 Mesh (90 µm)	15.0 max.
+325 Mesh (+45 µm)	80.0 min.

Sprayed Coating Properties

Bond Strength	17.2 MPa (2500 psi) min.
Hardness	Rockwell 15N, 75 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 20% max. Unmelted Particles - 25% max.

AMSC N/A

FSC 3439

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Deposit Efficiency and Deposition Rate

60% min for N_2/H_2 @ 14.0 lb/hr
50% min for A/H_2 @ 10.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF39
Pratt & Whitney Aircraft	PWA 1304
Allison Gas Turbine	EMS 56758

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/33)

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